Voronoi Space-Value Model:
Formulaic Implementation in High School Facility

Hong, Justin and Jung, Raymond
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Abstract

HS30322 홍승우 HS30313 정래윤 수학과제탐구
최종보고서

'보로노이 공간 가치 평가 모델: 고등학교 공간에서의 도식적 적용'은 콜레라 원인 검증에 활용된 보로노이 다이어그램의 문제 해결 방식에 영감을 받아 제작한 접근성 기반 공간 가치 평가 도식을 설명한다. 학생 설문과 정래윤 연구원 개인 설문을 기반으로 도출한 두개의 도식을 균등한 거리로 나누어진 청심국제고등학교 도면의 14×12 그리드에 대입하여 점수를 산출하고, 이를 색상 구배로 시각화하여 학교 건물 1 층에서 5 층 사이에서 가장 가치가 높은 공간의 위치를 도출했다. 본 연구에서 조사한 간단한 거리 기반 인수를 활용한 도식과 시각 다이어그램은 거시세계에 접목하여 복합 건물 또는 도심지의 실수요를 시각화하고, 이를 기존 부동산 지표와 비교하여 미래 공간 수요를 예측하는데 활용될 수 있다.

Introduction

Mathematical models and diagrams help humans better understand human decisions and the social relationships among individuals committing such actions—or in other words, social science.[1] One such case is that of finding the reason for the 1854 London Cholera outbreak, where physician John Snow implemented a simple geometric diagram to substantiate his theory that the epidemic was waterborne. By outlining human traffic on the assumption that all residents in the disease-stricken district procure water from the nearest communal well[Fig.1], Snow was able to pin-point the contaminated water source, thereby proving the liquid-borne nature of Cholera.[2] What Snow drew is known as a simple Voronoi Diagram.

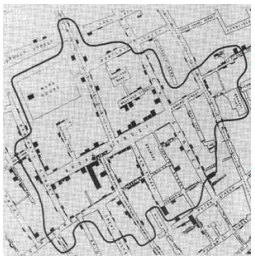


Fig. 1. Snow's Map[2]

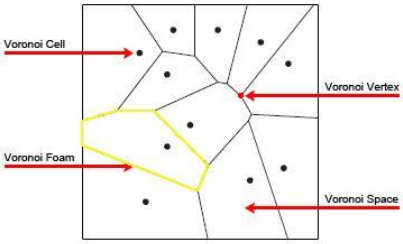


Fig. 2. Voronoi Diagram Form[3]

Voronoi diagram (Voronoi Tessellation), named after mathematician Georgy Voronoi, is a "decomposition-specified metric space with distance to a particular set of discrete objects in space, such as a set of discrete points."[3] By dividing a planar surface into a subset represented by area that is nearest to a set point on the surface, the diagram provides an exhaustive wayfinding fabric. A typical Voronoi Diagram consists of Voronoi cells, foams, spaces, and vertexes.[Fig.2]

Inspired by the wayfinding nature of Voronoi diagrams that Snow utilized, this study pursued implementation of its concept to quantify a numeric space value assigned to a set of exhaustive grids within a controllable space—the school facility. The wayfinding nature of Voronoi tessellation limits each of its outputs segmental, meaning that one entity in a strip of space (a residence in Snow's case of Cholera outbreak) is bound to commit to a single source point (the nearest well). To overcome this restraint, this study utilized a gradient space-value distance model, which incorporates vector parameters of x, y, and z and weight of spatial significance as input and absolute-scale points ranging from zero to 20,000 as output. The study then visualized the point-scales into color-coded diagrams to intuitively distinguish the hierarchy of desired intramural space based on accessibility.

Mathematical Model

Grid

Vectors (m)	x	y
Elevator	100	55
Mid. Staircase	75	50
Side. Staircase	95	50
Cafe. Staircase	160	67
Canteen	156	73
Cafeteria	165	67
Gymnasium	150	46
Reading Room	66	50
Classroom	83	74

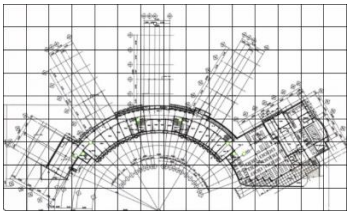


Fig. 3. Vector Locales of Significant Facilities

Figure 3 is a grid sample of a selected floor of the school facility. By utilizing CAD (Computer-Aided Design) software Rhinoceros 3D, the study pin-

[1] Nisbet, "social science."
[2] Thomas, "Uncovering the cause of cholera."
[3] Bahraminejad and Babaki, "Application of Voronoi," 1777

pointed the exact vector data of significant locales in the school facility that the student body is incentivized to visit. The study concluded 14×12 grid with each grid sized 13m in both width and length as an ideal and exhaustive representative and a visual model to distinguish quantitative spatial significance of this particular building.

The unit of measurement was set in meters. The origin (0,0) of xy-plane of each flower lower left of the exhaustive grid. The study pin-pointed the exact locales of significant school facilities as xy-vector coordinates, which are taken as quantitative input.

Weight of Spatial Significance

#/week	Raymond	Average
Elevator	25	20
Mid. Staircase	5	10
Side. Staircase	7	10
Café. Staircase	5	5
Canteen	3	4
Cafeteria	9	14
Gymnasium	11	2
Reading Room	2	1
Classroom	9	15

Fig. 4. Survey Data of number of visits

While the school building houses floors ranging from B1F[Fig.3] to 5F, the study excluded B1F from the final visualized diagram as confounding variables such as traffic inconsistencies and unidentifiable traffic movement particular to the sector deters the study's goal to provide an intuitive visual model for generally desirable areas in the school facility. However, significant facilities housed within in B1F including the Cafeteria and Canteen remained as a point factor that affects the weighted space-value formula.

The study randomly sampled 50 members of student population (over 10% of total student body) the number of times each member visits a significant school facility per week mentioned in the table above. The numbers are weighted and implemented within the vector formula written below.

Formula

The vector distance can be represented as followed:

$$d(p,l) = x(p) + y(p) + z(l) \quad \text{(Eq. 1)}$$

where p is equivalent to the parameter vector locale of each grid and l is the distance represented by moving between floors. Computing the two parameters, the formulas of distance from each vector is as followed:

$$x(p) = |100-p| * 20 + |75-p| * 10 + |95-p| * 10 + |160-p| * 5 + |156-p| * 4 + |165-p| * 14 + |150-p| * 2 + |66-p| * 1 + |83-p| * 15 \quad \text{(Eq. 1a)}$$

$$y(p) = |55-p| * 20 + |50-p| * 10 + |50-p| * 10 + |67-p| * 5 + |73-p| * 4 + |67-p| * 14 + |46-p| * 2 + |50-p| * 1 + |74-p| * 15 \quad \text{(Eq. 1b)}$$

$$z(l) = (|0-l| * 18 + |1-l| * 2 + |2-l| * 1 + |5-p| * 15) * 62 \quad \text{(Eq. 1c)}$$

The difference in coordinate values was multiplied by the frequency of investigation, which was then stored as functions for each axis. The sum of the vector functions derives Eq.1.

In addition, the argument was set to enter a corresponding value according to the different coordinates in the grid to obtain the *total distance that must be moved according to the frequency at that location* for all 1,400 coordinates. The first half of 1,400 coordinates is derived from the general functions written above, and the second were derived by tweaking the function from the weighted frequency of a surveyed average student to the that of researcher Raymond Jung conducting the study.

Inter-floor distance was measured based on the average calorie consumption taking stairs and that of walking.[4] By corresponding the energy consumed with the length of trip taking the stairs, the study calculated the amount of resource consumed to take inclined steps as equivalent to that of walking 63m in flat plane. This factor is implemented in the formula as a coefficient of frequency.

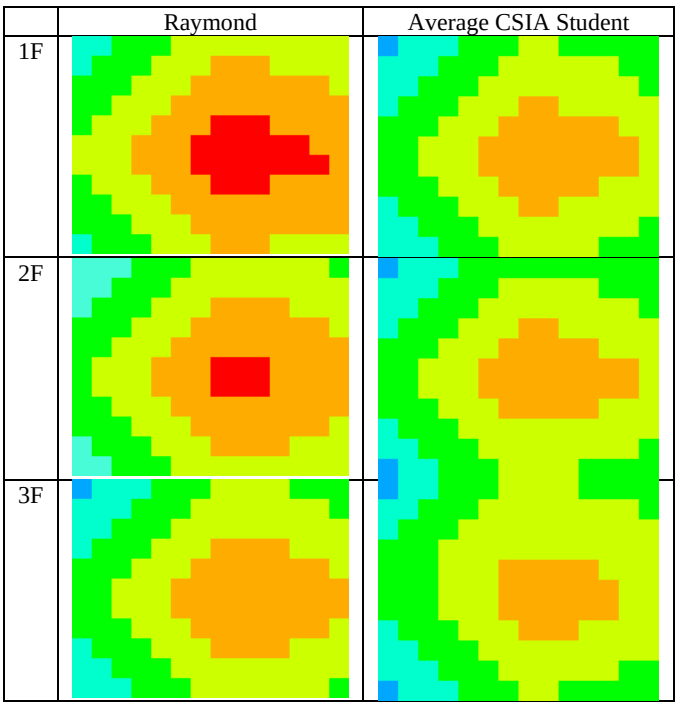
[4] Bumgardner, "Take the Stairs to Stay Fit and Healthy."

Implementation

Fig. 5. Fig. 6. Visual Model of Score Distribution

The gradient color ramp[Fig.5], reassembled from an existing color palette[5], is arranged hierarchal from red (most accessible) to blue (least accessible). Higher absolute score indicates greater distance from points of significant locations, implying less general accessibility for a surveyed individual.

Figure 6 depicts the absolute score distribution of researcher Raymond's model of 1st floor of school facility. Using conditional formatting script, the gradient is bounded hierarchically.



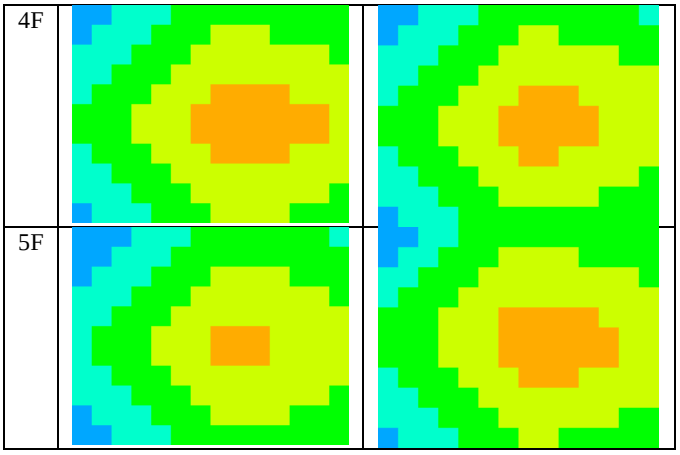


Fig. 7. Simplified Space-Value Gradient Map

The simplified color gradient model of spatial accessibility of two survey models Raymond and Average CSIA student[Fig.7] is depicted above.

Through a color-coded gradient diagram, the readers are able to intuitively distinguish the significance level—desirability—of particular grid space of school facility for surveyed individual Raymond and a model of an average CSIA student calculated from a sample of 50. The readers are able to note exceptionally high spatial desirability above (Absolute score below 4K) in the middle region of 1F and 2F of Raymond's model, which is a direct representation of the respondent's much frequent visit in the Gymnasium located in 1F compared to the average student model.

Limitations & Conclusion

The study assumed spatial accessibility as equivalent to desirability, yet numerous[6] [7] [8] case studies and research suggest that multiple confounding variables such as natural environment and space composition significantly affect human desirability of urban spaces. Nonetheless, Shokri[7] indicated that distance is an influential factor of spatial desirability based on the opinion of architectural theorists.

The space-value distance model can be utilized in a macroscale context: multi-purpose commercial plazas and metropolises can be evaluated through a single-factor visual diagram. The set of new data can serve to note a discrepancy between real estate

[5] Dassault Systèmes, "Color Ramp"
[6] Heydari *et al.*, "Explaining the Desirability Components"
[7] Shokri *et al.*, "Assessment of the Visual Qualities"
[8] Karimi *et al.*, "Effective Factors on Desirability of Private Open Spaces"

demand and accessibility, which could potentially hint under or overrated sectors in the speculative real estate market.[9] Such is of great public interest in South Korea, directly mentioned as a second to most important agenda of two largest parties in the bipolar congress.[10] *Voronoi Space-Value Model: Formulaic Implementation in High School Facility* is a step towards generalizing and quantifying the real estate market, a necessary demystifying process to ensure public financial and knowledge equity.

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[9] 하종숙, “하향안정 국면 돌입한 서울 부동산”

[10] 중앙선거관리위원회, “전국동시지방선거 정당정책”